

FinData

UPI Fraud Intelligence & Regulatory Surveillance Platform

An enterprise-grade platform transforming messy, disconnected payment logs into verifiable fraud ring intelligence, multivariate risk models, and grounded agentic AI.

20,000

TRUSTED
TRANSACTIONS

2,800

VALIDATED DISPUTES

295

DETECTED FRAUD RINGS

42/42

AUTOMATED TESTS
PASSED

Real-World UPI Telemetry Chaos

Raw payment data is corrupted, disconnected, and weaponized by fraud syndicates

The Raw Source Data Breakdown

- **20,400 Raw UPI Transactions:** Spaced UTRs (1,881), negative refund amounts (429), 14 status casing variants, and 400 exact row duplicates.
- **36,400 Raw Customer KYC Records:** 4,235 duplicate IDs with conflicting PANs, Aadhaars, and KYC statuses across multiple rows.
- **6,210 Raw Merchant Masters:** 1,127 duplicate IDs with conflicting bank accounts, missing MCCs, and trade name discrepancies.
- **2,884 Raw Chargeback Logs:** 219 unlinked orphan disputes and 183 empty disputed amount values.

Why Generic Pipelines Fail

The Naïve Approach: Generic pipelines either blindly drop dirty rows (distorting financial volume) or run unverified dashboards on corrupted identifiers.

The FinData Approach: We enforce *Preservation over Deletion*. Every anomaly is standardized, scored with resolution confidence, linked via foreign keys, and cataloged into an immutable audit diff matrix.

The Syndicate Reality: Fraudsters exploit multi-entity setups —319 shared Aadhaar clusters and 71 shared settlement bank accounts connect seemingly independent accounts into organized rings.

Forensic Entity Resolution & Reconciliation Diff

Multi-stage deterministic standardizers preserving 100% data auditability

36.4k → 28.9k

GOLDEN CUSTOMERS RESCUED

Hierarchy-based scoring resolving 4,235 duplicate IDs. Priority given to verified KYC status, valid PAN checksums, and 12-digit Aadhaar formats.

6.2k → 4.3k

GOLDEN MERCHANTS RESCUED

Consolidated 1,127 merchant ID conflicts. Rescued trade names, canonical categories, normalized MCCs, and active settlement bank accounts.

20.4k → 20.0k

TRUSTED UPI TRANSACTIONS

Removed 400 exact row duplicates. Classified negative amounts as refunds, stripped UTR spacing, and linked customer and merchant foreign keys.

Live Benchmark Curated Case: MCH7912 (6-Row Conflict)

Raw merchant arrived across 6 conflicting rows with two different bank accounts (ZNNS . . . vs XOLJ . . .) and divergent statuses (*Active*, *Enabled*, *Suspended*). Our engine selected the highest-fidelity active record with valid settlement banking and mapped all 6 raw records in an interactive side-by-side audit matrix.

Merchant Category Chargeback Ratio Distribution

Dynamically computed from 20,000 trusted transactions and 2,800 validated chargebacks

RANK	MERCHANT CATEGORY	TXNS	CBS	DISPUTE RATIO
#1	Apparel	145	44	30.34%
#2	Miscellaneous Retail	1,632	308	18.87%
#3	Department Store	173	31	17.92%
#4	Transportation	2,975	443	14.89%
#5	Telecom	126	18	14.29%
#6	Restaurant	3,000	419	13.97%
#7	Grocery	5,840	767	13.13%
#8	Pharmacy	3,004	385	12.82%
#9	Hotel & Lodging	2,964	369	12.45%
#10	Books & Stationery	141	16	11.35%

Benchmark Executive Insights

Highest Dispute Rate: Apparel (30.34%) exhibits the platform's highest dispute density—more than double the platform average of 14.00%.

Highest Absolute Loss Volume: Grocery records ₹2.66M disputed across 767 chargebacks, followed by **Transportation** (₹1.38M) and **Restaurant** (₹1.36M).

Highest Total Chargeback Count: Grocery (767) and **Transportation** (443).

Policy Takeaway: Apparel requires immediate mandatory delivery telemetry verification and lower chargeback tolerance thresholds.

Multi-Hop Fraud Rings & Syndicate Topology

Uncovering 295 coordinated networks across 3 distinct fraud typologies



Shared Settlement Mules

71 Rings

Multiple merchant entities routing settlement revenue into single bank accounts. E.g., RING-SETTLE-001 links 6 storefronts funneling into account XXXX0207.



Bust-Out Merchants

184 Rings

Commercial storefronts with rapid ticket size escalation (>2.5x declared baseline) followed by extreme chargeback surges before account abandonment.



Synthetic Identity Clusters

40 Rings

Multiple customer profiles sharing single Aadhaar/PAN credentials conducting circular money flows and coordinated repeat disputes.

Interactive Canvas Network Visualizer

Investigators explore multi-hop connections with physics layouts, interactive drag-and-drop node inspection, typology filtering, and 1-click generation of statutory investigation dossiers.

Multivariate Merchant & Customer Risk Models

0–100 composite risk scoring combining transaction anomalies and network topology



Merchant Risk Index (0–100)

- **Dispute Rate vs Category Cap (35 pts):** Penalizes excessive chargeback velocity.
- **Gross Disputed Volume (25 pts):** Evaluates total financial exposure.
- **Ticket Size Anomaly (20 pts):** Compares actual ticket size against declared onboarding baseline.
- **Shared Mule Account Flag (20 pts):** Immediate trigger if banking account is shared.

Top Merchant Flagged: MCH9291 (Score: 95.0, 78 disputes, 29.5% dispute rate).



Customer & Identity Risk Index (0–100)

- **Repeat Dispute Abuse (40 pts):** Flags accounts with >3 disputes or >50% dispute-to-transaction ratio.
- **Shared Aadhaar Cluster (25 pts):** Identifies synthetic identity syndicates (319 clusters).
- **Conflicting KYC Record (20 pts):** Discrepancies across duplicate KYC submissions.
- **Rejected KYC Status (15 pts):** Unverified or rejected onboarding telemetry.

Top Customer Flagged: USR97772 (16 chargebacks across 18 transactions).

Risk Policy Simulator & Regulatory STR Dossiers

Interactive 'What-If' threshold modeling and standardized FIU-IND pre-filing packages



Interactive Risk Policy Simulator

Risk officers can adjust critical portfolio parameters dynamically:

- **Dispute Rate Cap:** 5% to 50% (Default: 20%)
- **Ticket Size Multiplier:** 1.0x to 4.0x (Default: 2.0x)
- **Mule Sharing Threshold:** 1 to 5+ merchants (Default: 2)

Real-Time Delta: Calculates exact portfolio impact—merchants flagged, dispute volume contained, and total platform dispute capture rate.



Regulatory STR Investigation Dossiers

Aligned with **PMLA Section 12** & **FIU-IND FINnet 2.0** guidelines:

- **Automated Dossiers:** Generated for all 295 rings and high-risk merchants.
- **Complete Evidence Trail:** Itemized transaction logs with UTRs, timestamps, and amounts.
- **Dual Export:** Formatted Legal Markdown and Structured JSON for machine ingestion.

Domain-Guarded Grounded Agentic AI

Zero-hallucination factual extraction powered by OpenRouter Free Tier & Deterministic Fallback

🛡️ Multi-Tier Security & Fallback Pipeline

[User Prompt]
↓
[Domain Guard] → Intercepts Out-of-Domain & Injections
↓
[Deterministic Engine] → Extracts Verified Facts from Data
↓
[OpenRouter Free Tier] → Finance LLM Synthesizes Executive Brief
↓
[Deterministic Fallback] → 0ms Failover on Upstream Outages

⚡ Live Benchmark Verification Highlights

- **Benchmark Query:** Correctly answers "Apparel — 30.34%" with full comparative category context.
- **Phrasing Invariance:** 4/4 semantic variations map to canonical analytical facts.
- **Out-of-Domain Refusal:** 100% blocked at application level.
- **Prompt Injection Defense:** System overrides and DAN attacks safely neutralized.
- **Credential Safety:** Zero API keys exposed in client responses or git history.

Engineering Excellence & Test Verification

100% test pass rate across unit, integration, and browser automation suites

42 / 42

PYTEST TESTS PASSED

Unit, standardizer, risk engine,
and agent tests

13 / 13

SMOKE TESTS PASSED

End-to-end API and SPA
verification

8 / 8

PLAYWRIGHT VISUALS

1920x1080 automated visual
capture proofs

0 ms

OUTAGE DOWNTIME

Deterministic fallback on total
LLM failure

Modern Enterprise Tech Stack

PYTHON 3.12

FASTAPI

PANDAS & NUMPY

REACT 19 & TYPESCRIPT

TAILWIND CSS

RECHARTS & CANVAS GRAPH

PLAYWRIGHT AUTOMATION

OPENROUTER FREE TIER

From Unreliable Payment Data to Auditable Intelligence

"FinData doesn't just display data—it transforms raw chaos into an auditable, regulatory-aligned fraud intelligence platform that actively helps investigators detect, understand, and contain financial crime."

Key Competition Deliverables Ready:

- **Public GitHub Repository:** Clean git history, full documentation, zero secret leaks.
- **Authoritative Ground Truth:** Single source of truth across all 20,000 transactions and 2,800 chargebacks.
- **Statutory STR Dossiers:** Automated pre-filing compliance packages aligned with FIU-IND guidelines.
- **Grounded Agentic AI:** Invariant benchmark answers (Apparel: 30.34%) with robust security guardrails.